THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET A

Chapters 7 & 8: **The Mathematics of Maybe: Introduction to Probability, Predicting What Comes Next: Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A fair die is rolled once. The probability of getting an even number is: 1

(a) $\frac{1}{4}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{2}{3}$

2. A fair die is rolled once. The probability of getting a factor of 12 is: 1

(a) $\frac{1}{6}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{5}{7}$ **(d)** $\frac{5}{6}$

3. A bag contains 7 red, 5 blue and 8 green balls. One ball is drawn at random. The probability that it is green is: 1

(a) $\frac{1}{5}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{2}{5}$ **(d)** $\frac{3}{5}$

4. A bag contains 10 red, 6 blue and 4 green balls. One ball is drawn at random. The probability that it is blue is: 1

(a) $\frac{3}{10}$ **(b)** $\frac{7}{10}$ **(c)** $\frac{3}{11}$ **(d)** $\frac{2}{5}$

5. A letter is picked at random from the word ‘MATHEMATICS’. The probability of picking the letter M is: 1

(a) $\frac{3}{11}$ **(b)** $\frac{2}{11}$ **(c)** $\frac{9}{11}$ **(d)** $\frac{1}{6}$

6. For the sequence $t_n = 5n - 3$, the value of t_{10} is: 1

(a) 94 **(b)** 47 **(c)** 46 **(d)** 48

7. Which term of the sequence $t_n = 5n - 3$ is 607? 1

(a) $n = 124$ **(b)** $n = 123$ **(c)** $n = 122$ **(d)** $n = 121$

8. Which term of the sequence $t_n = 7n + 2$ is 261? 1

(a) $n = 38$ **(b)** $n = 37$ **(c)** $n = 39$ **(d)** $n = 36$

9. The 25th term of the AP $2, 6, 10, \ldots$ is: 1

(a) 196 **(b)** 97 **(c)** 99 **(d)** 98

10. The sum of the first 20 natural numbers is: 1

(a) 209 **(b)** 210 **(c)** 211 **(d)** 420

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. If the probability that a team wins is 0.35, what is the probability that it does not win? Why? 2

12. A card is drawn from a well-shuffled pack of 52 cards. Find the probability that it is 2

(i) a king

(ii) a red king.

13. Find the 10th and 26th terms of the AP $3, 8, 13, 18, \ldots$ 2

14. Which term of the AP $21, 18, 15, \ldots$ is -81? Is 0 a term? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A bag of 12 balls has red ones with $P(\text{red}) = \frac{1}{3}$. 3
(i) How many red balls?
(ii) If 2 more red balls are added, find the new $P(\text{red})$.

16. Which term of the GP $2, 6, 18, \ldots$ is 486? 3

17. Distinguish between subjective and objective (experimental/theoretical) probability using the example ‘it will rain today’. 3

OR

A spinner has 8 equal sectors numbered 1 to 8. Find the probability that it stops on 3
(i) a prime
(ii) a number greater than 5.

18. A ball dropped from 24 ft rebounds to $\frac{3}{4}$ of its previous height each time. Find the heights after the first three bounces and state the type of sequence. 3

OR

In the Sierpiński triangle the black area at Stage 0 is 1 sq. unit. Find the black area at Stage 3 and show it equals (number of black triangles) × (area of each). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Three coins are tossed together. 5
(i) List the sample space.
(ii) Find the probability of exactly two heads.
(iii) Find the probability of at least one head.
(iv) Find the probability that all three coins show the same face.

OR

Case study: In a factory, 8 of a sample of 200 bulbs are defective. 5
(i) Find the probability that a bulb is defective.
(ii) Estimate the number of defective bulbs in a batch of 5000.
(iii) Find the probability that a bulb is good.
(iv) Why would a bigger sample give a more reliable estimate?

20. Classify each as AP, GP or neither, and give the nth term where it exists: 5
(a) $7, 11, 15, 19, \ldots$
(b) $2, 6, 18, 54, \ldots$
(c) $1, 4, 9, 16, \ldots$
(d) $100, 50, 25, 12.5, \ldots$

OR

A sequence has $t_1 = -5$ and $t_{n+1} = t_n + 3$. 5
(i) Write the first five terms.
(ii) Find the explicit rule.
(iii) Is 52 a term?
(iv) Find the sum of the first 20 terms.

END OF QUESTION PAPER — **Total: 40 Marks**

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